
Measuring all the noises of LLM Evals

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Abstract

Separating signal from noise is central to experimental science. Applying well-established statistical method effectively to LLM evals requires consideration of their unique noise characteristics. We clearly define and measure three types of noise: prediction noise from generating different answers on a given question, data noise from sampling questions, and their combined total noise following the law of total variance. To emphasize relative comparisons and gain statistical power, we propose the all-pairs paired method, which applies the paired analysis to all pairs of LLMs and measures all the noise components based on millions of question-level predictions across many evals and settings. Our sample estimators are validated extensively on generative models and shown to be consistent with the bootstrap and sign test. We highlight two findings. First, each eval exhibits a characteristic and highly predictable total noise level across all model pairs. Second, paired prediction noise typically exceeds paired data noise, which means reducing prediction noise by averaging can significantly increase statistical power. These findings enable practitioners to assess significance without custom testing and to detect much smaller effects in controlled experiments.

1 Introduction

Separating signal from noise is central to experimental science. Well-established methods from statistics *can increase the rigor of the conclusions drawn from data* (Benjamini et al., 2021). Even though these methods are in landmark textbooks and clearly described for LLM evals (Miller, 2024), applying them effectively still requires some consideration of the prediction noise uniquely important for rich generative models like LLMs. Otherwise much potential power is lost (Example 1) and confusions ensue. Unlike previous probabilistic models that typically have a clear best prediction, LLMs produce persistently diverse predictions. Unlike physical experiments where treatments cannot be undone, digital LLMs allow for repeated independent samples, thus allowing us to directly measure their prediction noise. To better understand the sources of noise, we decompose the total variance into two interpretable components using the law of total variance. For a model A evaluated on questions x with prediction seed ϵ , we have

$$\begin{aligned}\text{Var}_{x,\epsilon}[A(x,\epsilon)] &= \text{Var}_x[\text{E}_\epsilon[A]] + \text{E}_x[\text{Var}_\epsilon[A]], \\ \text{Total variance} &= \text{data variance} + \text{prediction variance}.\end{aligned}$$

The *prediction variance* $\text{E}_x[\text{Var}_\epsilon[A]]$ is due to the model generating different answers on a fixed question. The *data variance* $\text{Var}_x[\text{E}_\epsilon[A]]$ measures how the questions vary in difficulty level and thus vary in their expected metrics. Prediction variance can be reduced but usually remains persistent; Data variance matters for reliability beyond the specific question set; The total variance is what matters in an experiment and is available even without multiple metrics per question.

While prediction variance can be reduced by averaging, the data variance is irreducible and can be large when the questions have different difficult levels. However, similar models may have similar performance on each question, suggesting that greatly reduced data noise is only possible with the paired analysis. A difficulty is that the paired method only applies to pairs of models rather than a single model or a list of models. In this work, we propose the all-pairs paired method to measure all the noises, which use the paired analysis on all pairs of LLMs to measure the data, prediction,

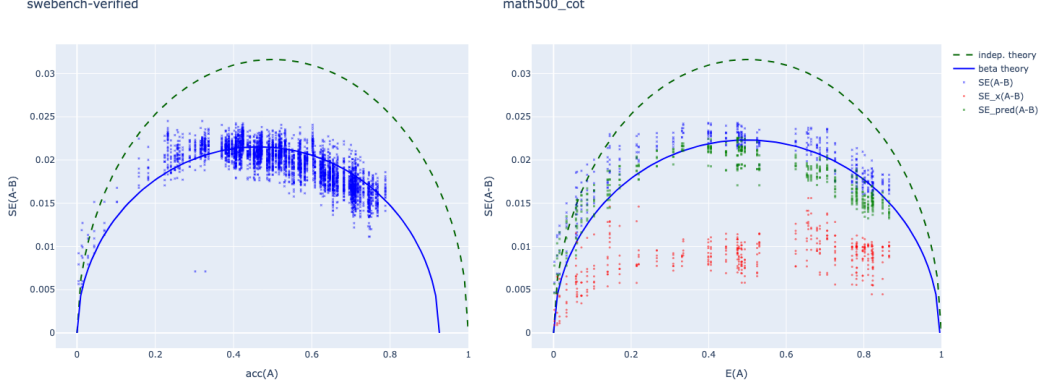


Figure 1: paired total standard errors vs. the accuracy, showing a clear trend in the empirical results agreeing with the beta theory prediction. Left: on SWEbench-verified 1 prediction per question so only the total noise is estimated, Right: MATH500 with 1000 predictions per question and estimated data noise SE_x , the prediction noise SE_{pred} , and total noise SE . More details in §4.

and total paired variances $\text{Var}[A - B]$, for all model pairs A, B , and across many evals and settings, revealing clear patterns.

We highlight two findings and their applications. First, each eval has a characteristic total noise level that holds across all pairs of LLMs or agents. Specifically, the paired test usually gives a total paired variance similar to the basic variance predicted by the accuracy p_A of model A , resulting in this rule of thumb on correctness evals

$$\begin{aligned}\text{Var}[A - B] &\approx \text{Var}[A] = p_A(1 - p_A), \\ \text{SE}[A - B] &\approx \text{SE}[A] = N^{-1/2}\text{Var}[A]^{1/2}.\end{aligned}$$

To be more accurate, Figure 1 Left shows the paired total noise on SWEbench-Verified against our theoretical prediction. This allows us to have accurate-enough noise measurements without custom analysis and even to infer the likely significance of results reported by others.

Second, by clearly separating prediction noise from data noise, we find that the paired prediction noise is typically greater than paired the data noise. For example, Figure 1 Right shows that the prediction noise is around two times the data noise on MATH500, which means that reducing the prediction noise can lead to a significant increase in statistical power, or a significant decrease in the minimum detectable effect size (Example 1). Even greater reductions can be expected for controlled experiments between close model pairs. When this condition holds, inferring the noise from the training curve becomes valid and useful (Example 2).

We speculate that the uniformity of these results come from the relative similarity among LLMs, since they are trained on similar data using similar methods. Like models, eval questions also have a global identity analogous to the ability to repeat experiments on the same subjects anywhere. Together, this allowed a decoupling of the noise measurements from particular experiments. Even in settings where these general patterns do not hold, our method is still applicable as long as the prediction noise is important and directly measurable.

Contributions. We clearly define and measure prediction noise, data noise, and total noise on many models and evals. Our general findings enable practitioners to assess significance without custom statistical testing, and often allow us to detect much smaller effects.

Contents. We present the main concepts informally and through examples in §2; then more precisely in §3. To ensure correctness, we validate our method by comparing with the well-established bootstrap and sign test methods (§3.5), and test our sample estimators (§3.4) against several generative models as well as on common LLM evals. §4 describes exploratory data visualization methods to gain insights into the data. In §6, we discuss why we observe these results and some caveats; why harder questions cannot yet compensate for smaller sample sizes; and provide some recommendations for practitioners to better deal with noise §6.4.

The data and analysis can be found at <https://all-the-noises.github.io>, which shows the noise component curves, reference noise values, etc across evals, temperatures, based on millions of question-level metrics from public leaderboards and controlled experiments.

2 Main concepts and examples

2.1 Types of noise

We should clearly distinguish three types of noise: the prediction sampling noise when answering a given question, the data sampling noise due to using a finite sample from the population of possible questions, and their sum total noise.

Prediction noise. The prediction noise comes from generating different answers on a given question with the same model. LLMs have the remarkable ability to sample independent predictions that are especially diverse on reasoning and agentic questions. In the physical world, the prediction noise typically cannot be measured directly — humans cannot completely forget their previous thoughts and medical treatments cannot be undone to get more independent samples. Previous probabilistic ML models such as classifiers can typically make their best prediction directly and do not have interesting prediction noise.

The prediction noise is also persistent even though we can try to reduce it by fixing the random seed, decreasing the temperature, averaging over many samples per question, or applying an aggregation method such as majority voting and verification. Fixing random seed or lowering the temperature is brittle, which only works on the exact same model. Trying to produce the best answer by aggregation is perhaps the most sound method, though this still leaves some prediction noise due to the diversity of samples. Among these noise reductions, only averaging and fixing the random seed gives the same mean while others change the mean as well. Averaging is the most general and effective, which we will focus on.

On top of sampling, additional prediction noise can come from inference-time choices such as prompt formatting, hyperparameters, and answer extraction; and training-time choices such as the random seeds, hyperparameters, and training steps. All of these can theoretically be measured by repeating the process of changing the choice, optionally training the model, and then drawing predictions on each question. Training is not typically considered noise, but it may produce more noise than signal over a small amount of steps.

Data noise. The eval data sampling noise comes from sampling a particular set of N questions instead of another equally good N questions from the unobservable population of questions. As long as the model has different expected accuracy on different questions (i.e. some questions are harder), the mean will vary when a different set of questions is sampled. Unlike the prediction noise, the data noise cannot be directly measured by evaluating on the fixed, given set of questions; and it cannot be reduced. Though the data noise can be inferred using the bootstrap or the variance estimator, it is not observed in experiments such as the training curve (Example 2) and varying the training seeds (Madaan et al., 2024).

To see the presence of data noise, suppose that out of 100 True and False questions, model A is correct on 50 random questions, B is correct on 52 random questions out of the same 100 questions. Then B is not reliably better than A even when each question is answered deterministically with no prediction noise. Suppose we have $p = 0.5$ chance of sampling a question that is answered correctly by A , then the number of correctly answered questions has a standard deviation of $(100p(1 - p))^{1/2} = 5$, and thus it is typical to see a difference of more than 2 questions. Statistical methods such as computing the standard error or bootstrap focus on the data noise and will show that A is not reliably better in this case.

If we only care about a specific set of questions, then we can ignore the data noise. There are real life situations when the goal is to memorize a specific list of questions such as a driving knowledge test where the set of questions is published, or to solve specific important problems (AGI/ASI, P vs. NP). On most evals, we care about the underlying ability measured by the eval rather than the specific questions of the eval, so the data noise is not to be ignored.

Total noise = prediction + data noise. We can make this relation precise using the law of total variance. The extensive literature about noise focuses on the total noise or the data noise, perhaps

because the prediction noise is uninteresting as in classification, or not measurable directly as in most non-digital experiments.

Paired analysis. Whereas insights, skills, and resources are required to improve LLM ability, the paired methods can increase the signal significantly. Thus they should be recommended as enormous resources are devoted to improving models. The paired method makes use of prediction differences on the same question to gain more information. To see why the paired method is better on real data, Figure 4 shows the heatmap of correctness where models at a similar overall accuracy (close in x -axis) also do similarly on the individual questions. Thus, the paired method has the potential to reduce the data noise significantly. Examples 2b and 1 show the potential increase in power using paired methods, especially when prediction noise $>$ data noise, leading to the ability to detect differences several times smaller. Example 3 shows the basic idea of paired analysis.

Example 1 (HumanEval). On HumanEval with $N = 164$, if a model has $\bar{A} = 0.5$ accuracy, then the unpaired standard error (SE) is $\sqrt{1/164 \times 0.5(1 - 0.5)} = 4\%$. Since a factor of $\sqrt{2}$ is needed when comparing pairs of models, we need $4\% \times \sqrt{2} \times 1.96 = 11\%$ to get 0.05 p -value. The table below shows the reductions with paired and averaging, and how the combination has the most potential.

Test Type	Standard error	Difference for $p < 0.05$
Unpaired	6%	12%
Unpaired with averaging	4%	8%
Paired single prediction	4%	8%
Paired with averaging	1-2%	2-4%

Table 10 of Llama 3 (Grattafiori et al., 2024) reports the unpaired 95% confidence intervals of around 7%, thus many of the comparisons do not clear the $\sqrt{2} \times 7\%$ difference with their method.

Example 2. a) The training curve is invalid in theory. The *training curve* plots the eval result as a function of training steps and graphically shows the variance of eval results as a function of training steps (Figure 2, col 1). While separated training curves is intuitively necessary for reliable results, it is not sufficient nor necessary. The training curve only shows the prediction noise and completely ignores the data noise since it uses the same eval questions repeatedly. In theory, the data noise can be much higher than the prediction noise. For example, suppose the training data of models A and B consists of a random leak of the 100 eval questions, each included with probability 0.5. Suppose that B is 5% more contaminated than A just due to noise. Then B has a consistently higher training curve if both A and B are only correct on the eval questions leaked into their training set and are wrong on the rest. Both models eventually reaching a constant peak after seeing all the leaked questions. In this example, any statistical method considering the data noise will tell us that B is not better than A , which is true by design.

b) The training curve is valid when prediction noise is dominant. In practice, if the prediction noise dominates the data noise, then the training curve method is valid. Figure 2 shows increasingly powerful analysis on the training curve visualized by the bootstrap.

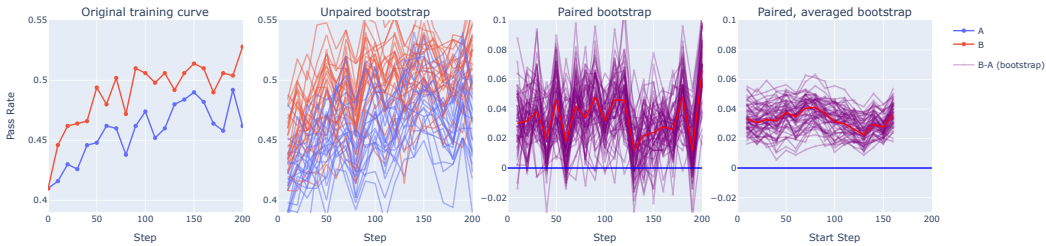


Figure 2: **Column 1) original training curve.** A and B both start from the same checkpoint at step 0 where data noise is 0. **2) unpaired bootstrap.** Many A s are higher than many B s, showing a meaningless difference between A and B using this analysis method. **3) paired bootstrap.** Slightly significant with a z -score of 1.7 at 1 prediction per question. **4) paired, averaged bootstrap.** The result is very significant with a z -score of 3.5 when the question level predictions are averaged over 5 consecutive training checkpoints to reduce the prediction noise.

Example 3 (Paired vs unpaired). In the unpaired case, suppose A scored 50% on this year’s exam, B scored 60% on last year’s exam where each exam contains different questions drawn from the same distribution. So B might have done better because last year’s exam contained easier questions. In the paired case, suppose B scored 60% on the same exam as A containing exactly the same questions, then the same 10% difference is more significant.

All-pairs paired method. We measure the paired standard error $\text{SE}(A - B)$ of all model pairs A and B , where the total noise is always measured, and the data and prediction noises are measured whenever possible. While this approach can be inconclusive in general (Example 4), our measurements show that on LLM evals this turned out to be quite informative, so we highlight the following two findings.

Finding: predictable total noise. Figure 1 shows the total noise vs. the accuracy of A . We observe that the total noise is quite predictable and agrees well with a theoretical model. This means each eval has a characteristic total noise as a function of the accuracy. This is the right noise number to use for experiments where one prediction is made per question.

Finding: typically prediction noise > data noise. When this holds, we can gain much statistical power by using the expected metric where the prediction noise is averaged away to be comparable to the data noise remaining. Figure 1 Right shows that on MATH500, the data standard error (SE) is typically less than $1/2$ of the prediction SE between different final models. Example 2b and Figure 2 show training curves evaluated on SWEbench-verified starting with the same checkpoint, the data SE is $\sim 1/6$ of the prediction SE. If we average over 36 predictions, then the total SE would be $\sqrt{2}/6 = 0.24$, allowing us to detect effects that are 0.24 the size with the same p -value and power. Example 1 shows that the smaller reduction on HumanEval is still by a factor of 2. For controlled experiments, this allows us to detect differences several times smaller at the same significance level and power. Important caveats are discussed in §6.3.

Example 4 (Error bars for leaderboards). Gu et al. (2024); Chiang et al. (2024) tried to put error bars on a leaderboard. To do this, they use a fixed baseline for paired comparison. However, this is not generally meaningful: let A be the baseline, B and B' are a pair of similar predictions very different from A , so $\text{Var}[A - B] \gg \text{Var}[B - B']$. This example shows it’s not in general possible to have one error bar per model, or per accuracy with the paired method. However, at least for the total noise of LLM evals, this work shows that there is almost no such surprises, thus we can compute meaningful error bars.

3 Method

3.1 Setup and notations

To make these concepts precise, suppose there are N eval questions $\{x_1, \dots, x_N\}$, which are typically text prompts. The model makes prediction $\hat{y}(x)$ and we get $\text{metric}(\hat{y}, x)$, typically 0 for incorrect and 1 for correct but can be a real number or an aggregate metric. Any information needed to compute the metric such as the ground truth answer or agent environments are included implicitly. We use $A(x) := \text{metric}(\hat{y}(x), x)$ to denote the metric function evaluating model A on the question x .

On an eval, we compute the average of all N questions to get the mean metric

$$\bar{A} := \frac{1}{N} \sum_{i=1}^N A(x_i) \approx \mathbb{E}_x[A(x)].$$

The estimated variance of A and standard error (SE) of the mean are respectively

$$\begin{aligned} \text{Var}_x[A(x)] &\approx \frac{1}{N} \sum_{i=1}^N (A(x_i) - \bar{A})^2, \\ \text{SE}(A, N) &= N^{-1/2} \cdot \text{Var}[A]^{1/2}, \end{aligned}$$

where the standard error is normalized to the right scale for comparison to the effect size. We just say *noise* when there is no need to distinguish the variance and the standard error. N should be large enough for estimating the variance accurately, so a bias correction on $\text{Var}_x[A]$ using $1/(N - 1)$ is not useful.

With models A and B , the score difference $\bar{A} - \bar{B}$ can be compared to $\text{SE}(A - B) = (\text{SE}(A)^2 + \text{SE}(B)^2)^{1/2}$ to determine if the difference is likely due to chance. While this is simple and correct, it is also unnecessarily weak (Example 1).

Paired comparison. Whereas innovation, insight, or compute is required to get real signal, tightening the analysis is easy with the paired methods and should be recommended generally. $A(x)$ and $B(x)$ are correlated when the same question is used to evaluate both $A(x)$ and $B(x)$. To be more powerful for free, we can use the paired variance

$$\text{Var}_x[A(x) - B(x)] = \text{Var}_x[A] + \text{Var}_x[B] - 2\text{Cov}_x[A(x), B(x)].$$

The Cov term may potentially reduce the paired variance to 0 when A and B are perfectly correlated.

From the standard errors we can obtain the z -score $z = \frac{\bar{A} - \bar{B}}{\text{SE}[A - B]}$. When $N > 100$ is moderately large, then the Central Limit Theorem applies and the z -scores can be converted to p -values. For example, $\Pr[|z| > 1] = 0.32$ is very weak, $\Pr[|z| > 5] < 10^{-6}$ is beyond doubt, and $\Pr[|z| > 1.96] = 0.05$ for a p -value of 0.05 is a reasonable conventional standard.

3.2 Sampling multiple predictions on each question

To capture this setting, we use ϵ for the random seed that generates the prediction, which is independent of the question x , thus allowing the concise notation $\text{Var}_x[\mathbb{E}_\epsilon[A]] = \text{Var}_x[E_{\epsilon|x}[A(x, \epsilon) | x]]$. As before, but now also averaging over the K samples for each question x , we consider the average score

$$\bar{A}_K = \frac{1}{N} \sum_{i=1}^N \frac{1}{K} \sum_{j=1}^K A(x_i, \epsilon_{ij}) \approx \mathbb{E}_{x, \epsilon}[A(x, \epsilon)],$$

The estimated variance is then

$$\text{Var}_{x, \epsilon}[A(x, \epsilon)] \approx \frac{1}{N} \sum_{i=1}^N \frac{1}{K} \sum_{j=1}^K (A(x_i, \epsilon_{ij}) - \bar{A})^2.$$

We have the law of total variance satisfied by the components

$$\text{Var}_{x, \epsilon}[A] = \text{Var}_x[\mathbb{E}_\epsilon[A]] + \mathbb{E}_x[\text{Var}_\epsilon[A]].$$

$\text{Var}_{x, \epsilon}[A]$ is the total variance of first drawing a question x from the pool of questions, and then drawing a sample answer for this question. $\text{Var}_x[\mathbb{E}_\epsilon[A]]$ is variance of the expected score $\mathbb{E}_\epsilon[A]$, which we call the data variance. $\mathbb{E}_x[\text{Var}_\epsilon[A]]$ is the variance due to sampling from the model, which we call the prediction variance. See §A.1 for why $\mathbb{E}_x[\text{Var}_\epsilon[A]]$ is the same as the prediction variance that can be directly measured on a fixed eval.

All noises metrics can be normalized to be their respective standard errors,

$$\begin{aligned} \text{SE}(A, N) &= N^{-1/2}(\text{Var}_{x, \epsilon}[A])^{1/2}, \\ \text{SE}_x(A, N) &= N^{-1/2}(\text{Var}_x[\mathbb{E}_\epsilon[A]])^{1/2}, \\ \text{SE}_{\text{pred}}(A, N) &= N^{-1/2}(\mathbb{E}_x[\text{Var}_\epsilon[A]])^{1/2}. \end{aligned}$$

Paired comparison. When we have two models A and B , we can consider the paired variance $\text{Var}_{x, \epsilon}[A(x, \epsilon) - B(x, \epsilon)]$, which also satisfies the law of total variance on $A - B$,

$$\text{Var}_{x, \epsilon}[A - B] = \text{Var}_x[\mathbb{E}_\epsilon[A - B]] + \mathbb{E}_x[\text{Var}_\epsilon[A - B]]. \quad (1)$$

3.3 Estimating from samples

While (1) allows us to estimate the variance components directly from K paired samples $A(x, \epsilon_j) - B(x, \epsilon_j)$ for $j = 1, \dots, K$, this is inaccurate since the seeds or predictions are interchangeable. We can use this to derive formulas for estimating from samples accurately, as if using all $K \times K'$ pairs of collected predictions $A(x, \epsilon_j) - B(x, \epsilon'_k)$ for $j = 1, \dots, K$ and $k \in 1, \dots, K'$. First, the prediction variance $\mathbb{E}_x[\text{Var}_\epsilon[A - B]]$ decomposes by independence on each question x ,

$$\begin{aligned} \text{Var}_\epsilon[A(x, \epsilon) - B(x, \epsilon')] &= \text{Var}_\epsilon[A(x, \epsilon)] + \text{Var}_\epsilon[B(x, \epsilon)], \\ \mathbb{E}_x[\text{Var}_\epsilon[A(x, \epsilon) - B(x, \epsilon')]] &= \mathbb{E}_x[\text{Var}_\epsilon[A(x, \epsilon)]] + \mathbb{E}_x[\text{Var}_\epsilon[B(x, \epsilon)']]. \end{aligned}$$

Next, the data variance $\text{Var}_x[\mathbb{E}_\epsilon[A - B]] = \text{Var}_x[\mathbb{E}_\epsilon[A] - \mathbb{E}_\epsilon[B]]$ decomposes by linearity of expectation. Finally, the total variance is slightly tricky, where

$$\begin{aligned}\text{Var}_{x,\epsilon}[A - B] &= \text{Var}_{x,\epsilon}[A] + \text{Var}_{x,\epsilon}[B] - 2\text{Cov}_{x,\epsilon}[A, B] \\ &= \text{Var}_{x,\epsilon}[A] + \text{Var}_{x,\epsilon}[B] - 2\text{Cov}_x[\mathbb{E}_\epsilon[A], \mathbb{E}_\epsilon[B]].\end{aligned}\quad (2)$$

The equality is by the law of total covariance, $\text{Cov}_{x,\epsilon}[A, B] = \text{Cov}_x[\mathbb{E}_\epsilon[A], \mathbb{E}_\epsilon[B]] + \mathbb{E}_x[\text{Cov}_\epsilon[A, B]]$ where the second term is 0 since different random predictions on a given x are independent.

Small K correction. To estimate $\text{Var}_x[\mathbb{E}_\epsilon[A]]$ from K sampled predictions per question, a correction from the direct estimator can significantly increase the accuracy. We describe the concept for the unpaired case for simplicity but it matters the most in the paired case. By the law of total variance on $\bar{A}_K(x, \epsilon) = \frac{1}{K} \sum_{j=1}^K A(x, \epsilon_j)$, we have

$$\begin{aligned}\text{Var}_{x,\epsilon}[\bar{A}_K] &= \text{Var}_x[\mathbb{E}_\epsilon[\bar{A}_K]] + \mathbb{E}_x[\text{Var}_\epsilon[\bar{A}_K]] \\ &= \text{Var}_x[\mathbb{E}_\epsilon[A]] + \frac{1}{K} \mathbb{E}_x[\text{Var}_\epsilon[A]] \\ \implies \text{Var}_x[\mathbb{E}_\epsilon[A]] &= \text{Var}_x[\bar{A}_K] - \frac{1}{K} \mathbb{E}_x[\text{Var}_\epsilon[A]] \\ &\approx \hat{\text{Var}}_x[\bar{A}_K] - \frac{1}{K-1} \hat{\mathbb{E}}_x \hat{\text{Var}}_\epsilon[\bar{A}_K]\end{aligned}$$

where the hats mean sample estimate. This correction for small K is important because we will average over N questions. If the estimate on each question is biased by $1/(K-1)$, the average error will still be off by the bias no matter how big N is. Whereas if the estimate is unbiased on each question, then the average error can decrease as N increases. Figure 3 shows this observation on MATH500, where the small- K correction is needed to reduce the error to an acceptable level even with 2000 bootstrapped questions. The high relative error is because the paired data noise is much smaller than the prediction noise. An instructive example is to consider when $\mathbb{E}_\epsilon[A - B] = 0$ but an estimate using K samples will not be 0 due to non-zero prediction noise.

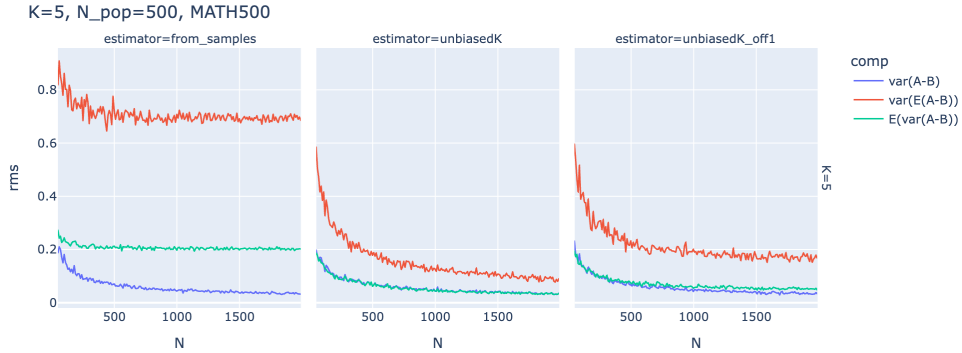


Figure 3: Relative errors of the paired variances. $K = 5$ samples are drawn from the 500 questions from MATH500, where 1000 real predictions are drawn from models A and B on each question, which is treated as the ground truth. The root mean squared relative errors (rms) of variance components are plotted vs. bootstrap sample size N , extending beyond the population size 500. The left figure without the correction on K has an unacceptably high 70% relative error even with 2000 data points. The middle figure is with the correction of §3.3 and the right figure uses $1/K$ instead of $1/(K-1)$.

3.4 Implementation in array notation

In an experiment, we typically evaluate the model on all N questions, and may draw K answers for each question. In this section, we overload our notation with actual samples $A, B \in \mathbb{R}^{N \times K}$ and present the formulas in §3.3 in numpy-style code. For simplicity, we use the same number of predictions $K_i = K$ on all questions x_i . For the unpaired case, we have Table 1. For paired, we have Table 2.

Name	Formula	Code	Bernoulli
total variance	$\text{Var}_{x,\epsilon}[A]$	<code>var(A)</code>	$\bar{p}(1 - \bar{p})$
data variance	$\text{Var}_x[\mathbb{E}_\epsilon[A]]$	<code>var(mean(A, axis=1))-b</code>	$\frac{1}{N} \sum_i (p_i - \bar{p})^2$
prediction variance	$\mathbb{E}_x[\text{Var}_\epsilon[A]]$	<code>mean(var(A, axis=1))+b</code>	$\frac{1}{N} \sum_i p_i(1 - p_i)$

where $b=1/(K-1)*\text{mean}(\text{var}(A, \text{axis}=1))$

Table 1: Unpaired estimators where $A \in \mathbb{R}^{N \times K}$ with K samples for each of N questions.

Formula	Code
$\text{Var}_{x,\epsilon}[A - B]$	<code>var(A)+var(B)-2*cov(mean(A,axis=1),mean(B,axis=1))</code>
$\text{Var}_x[\mathbb{E}_\epsilon[A - B]]$	<code>var(mean(A,axis=1)-mean(B,axis=1))-b</code>
$\mathbb{E}_x[\text{Var}_\epsilon[A - B]]$	<code>mean(var(A,axis=1)+var(B,axis=1)) +b</code>

where $b=1/(kA-1)*\text{mean}(\text{var}(A, \text{axis}=1))+1/(kB-1)*\text{mean}(\text{var}(B, \text{axis}=1))$

Table 2: Paired estimators where $A \in \mathbb{R}^{N \times K_A}$, $B \in \mathbb{R}^{N \times K_B}$.

3.5 Equivalence to the bootstrap and the sign test

Two other very well-established methods are the bootstrap (Efron and Tibshirani, 1986) and paired difference tests such as the sign test (Dixon and Mood, 1946). These methods ask if the observed results are likely when the questions are random for bootstrap, and when the comparison outcomes are random for the sign test. We show they all give the same answer.

Bootstrap is a general method where resampling the given questions uniformly with replacement is shown to give the right answer for many estimation problems, including estimating the standard error. That is, *resampling the given questions uniformly with replacement gives the same answer as getting more real questions from the population*. Applied to the problem of noise, the natural question is how likely we are to get the opposite result $\mathbb{E}[A] < \mathbb{E}[B]$ as opposed to the observed result $\mathbb{E}[A] > \mathbb{E}[B]$. If the opposite outcome also has significant probability, then the conclusion is not statistically significant.

The sign test predates bootstrap and instead of sampling the questions, it makes the comparison outcome random. It supposes that any difference actually happens with a random probability $\Pr[A(x) \neq B(x)] = 0.5$ (null hypothesis) and asks how likely we are to get a more extreme outcome than what we actually observe. This corresponds to a predictor that mixes the predictions of A and B with probability 0.5. If it is quite likely to observe a more extreme outcome under the null hypothesis or the mixed predictor, then the conclusion is not statistically significant.

The key step in both methods is estimating the variance of their respective random procedure. If the variances are the same, then the probability to observe the opposite result from an observed mean (bootstrap) is also the same as the probability of observing a more extreme result if the real mean is 0 (sign test). The details are deferred to §B.

4 Experiments

The experimental results of this paper can be found at <https://all-the-noises.github.io>, where the method is tested on more evals and where the figures are interactive to support more detailed investigations and case studies. We can use this to check that the main conclusion indeed holds on all the evals.

4.1 Exploratory analysis and findings

4.1.1 Data heatmap

To get an overview of our data, we use a heatmap shown in Figure 4. Each row is a different question, sorted from the easiest to the hardest. Each column is a model whose x -coordinate is the overall accuracy $\mathbb{E}[A]$ of this model. Like models, we might consider showing questions with y -coordinate at their average accuracy instead of rank, but that would depend on which models are included, and less intrinsic to the eval. We immediately observe that models at a similar overall accuracy also does similarly on individual questions. Around 1000 samples are drawn for each question and model for MATH500 in Figure 4. The SWEbench figure is based on the leaderboard data, which had to commit to single answers, but the same overall pattern holds, though with more noise. On MATH500, most of the red regions of bad models are still non-zero.

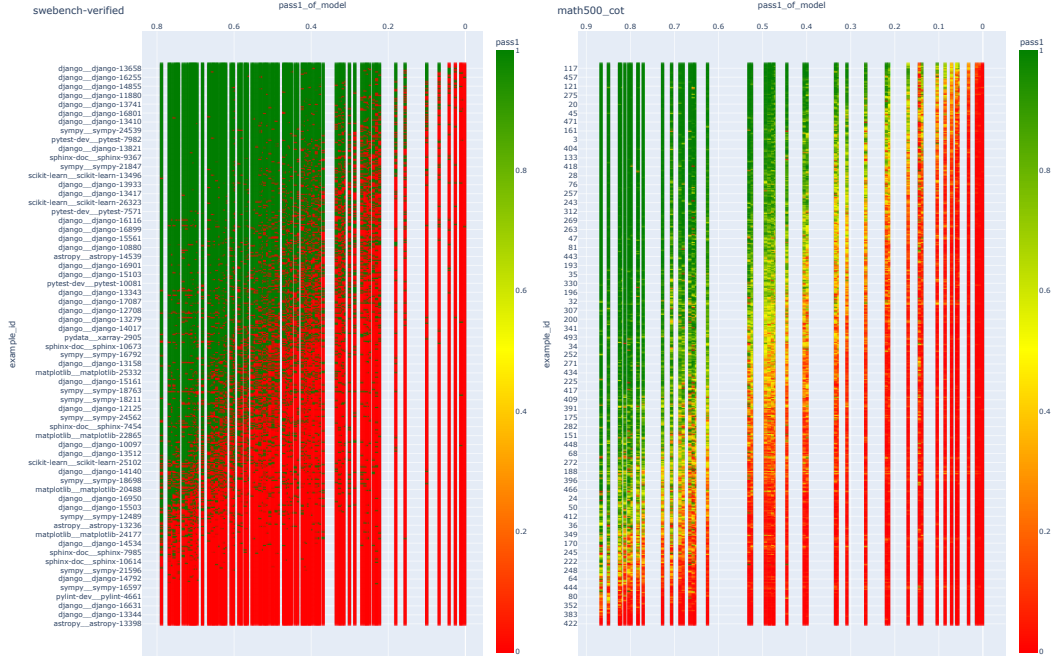


Figure 4: Results heatmap. Each row is a different question, sorted from the easiest to the hardest. Each column is a model whose x -coordinate is the overall accuracy. Left: SWEbench-verified, 1 prediction per question; Right: MATH500, 1000 predictions per question. See §4.1 for more details.

4.2 All-pairs paired noises and predictable total noise

Figure 1 shows that the total SE agrees well with the $\text{Beta}(p, 1 - p)$ theory predictions described in 4.2.1 and the range of the data noise and prediction noises. Only models pairs that are close in performance are plotted $E[A] - E[B] < 5 \text{SE}(A - B)$ to avoid pointless comparisons between models whose difference is not in doubt. In Figure 1, swebench-verified, we see 2 outlier results near $x = 0.3$ that comes from SWE-Fixer (Xie, 2025), due to using a deterministic filter instead of drawing another prediction from the model.

The noise components can be traded off so they can depend on the setting and not as predictable as the total. We find at more natural temperatures of $[0.7, 1]$, the prediction noise tend to be higher than the data noise across evals. Figure 5 shows the noise components at two different temperatures on CRUXEval, where the breakdown differs but the total noise remains the same.

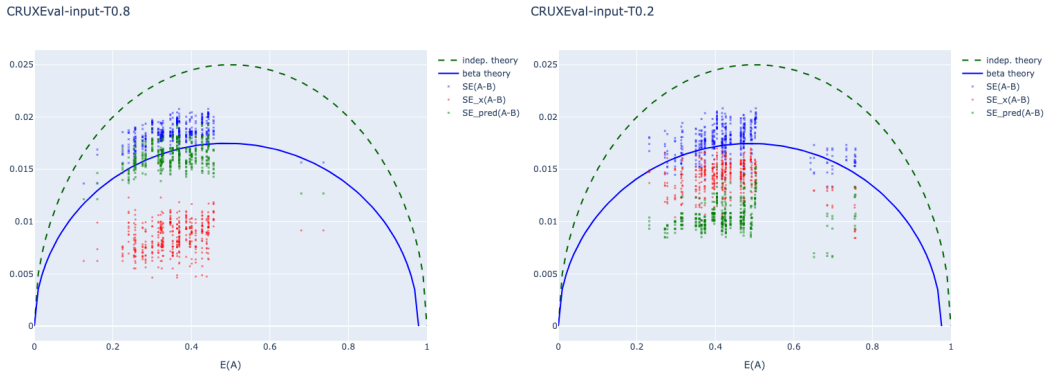


Figure 5: All noise components on CRUXEval at temperature 0.8 (left) and 0.2 (right). The prediction noise dominates at 0.8 temperature, whereas the data noise dominates at 0.2 temperature, while both still yielding about the same total noise.

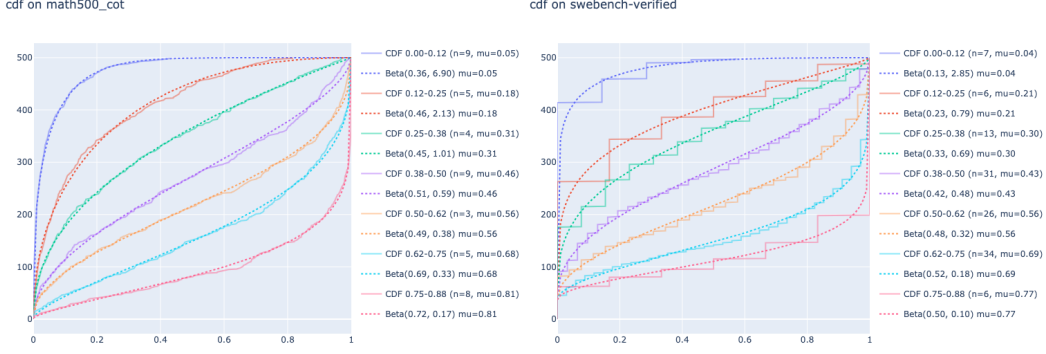


Figure 6: Empirical cumulative distribution curves (CDF, solid) and their respective Beta models (dots). Each color represents a different bin by model accuracy. The average accuracy of each question in each bin of models is computed and sorted to produce the CDF curves. The fit is good.

4.2.1 The Beta theory makes good noise predictions and fits the data

Figure 6 shows the the comparison between a $\text{Beta}(a, b)$ distribution and the empirical cumulative distribution (CDF) of the metrics, where the models are binned by range of accuracy. Then $\hat{p}_1 = c_1/K, \dots, \hat{p}_N = c_N/K$ is computed for each questions i across the K combined predictions among the binned models, c_i is the total number of predictions that evaluated as correct on question i . Smaller $a + b$ means more bimodal where $a + b$ tend to decrease when the model accuracy is higher, showing that better models become more bimodal whereas very bad models is usually unimodal. For an extreme example, guessing uniformly on multiple choice questions with C choices means the expected accuracy $u_i = 1/C$ for all questions, which is has a single mode at $1/C$.

The Beta theoretical model says that the expected accuracy of each question follows the $\text{Beta}(p, 1-p)$ distribution where p is the mean of the model. More precisely, the expected accuracy of question x_i is $u_i \sim \text{Beta}(p, 1-p)$ so that $A(x_i), B(x_i) \sim \text{Bernoulli}(u_i)$ and $E[A(x_i)] = u_i$. Too see that this leads to the observed SE, the models draw independent predictions on each x_i so $\text{Cov}_\epsilon[A, B | x_i] = 0$. Thus we have $E_x[\text{Var}_\epsilon[A - B]] = E_x[2\text{Var}_\epsilon[A]]$, which integrates to

$$\int_0^1 2u(1-u) \frac{u^{p-1}(1-u)^{(1-p)-1}}{B(p, 1-p)} du = \frac{B(p+1, (1-p)+1)}{B(p, (1-p))} = p(1-p),$$

where $B(a, b) = \int_0^1 u^{a-1}(1-u)^{b-1} du$ is the normalization constant for the $\text{Beta}(a, b)$ distribution. One consideration to fit the data better is that all models fail on some questions, so we suppose those questions actually have $u_i = 0$ instead of $\text{Beta}(p, 1-p)$.

4.3 Exceptions

We note a few exceptions to the general findings. On guessable evals such as MMLU, when the performance is near random chance, the observed noise is closer to the independent prediction rather than the Beta prediction. The predictions of Llama on vLLM is more deterministic than expected and thus has higher data noise than others. Distilled models tend to have less sharp distribution at the same performance level. These are relatively few and have good explanations.

5 Related work

Much work deals with noise in science and machine learning (Lehmann and Romano, 2005; Dror et al., 2018; Hermann et al., 2024; Card et al., 2020, among many others), we adopt the approach of Miller (2024) to estimate the variance directly. While using samples to estimate the population follows from the law of large numbers, their approach is clarifying, has unique considerations and recommendations for LLMs, and is directly generalizable all metric functions. Their method is also consistent with the bootstrap (Efron, 1979) and the sign test (Dixon and Mood, 1946) when applicable (§3.5). Efron and Tibshirani (1986) clearly described estimating the standard error as the example of not needing bootstrap. The concept of total variance decomposition is present the *analysis of variance*, but Section 3 of Miller (2024) applied this concept to LLMs under *variance of the conditional mean* and *mean conditional variance*, which we call the prediction noise and the

data noise. They also advocated for averaging (resample), but it is not clear how much gains can be expected. In this work, we develop and test the methods of estimating noise types from samples and then empirically measure these variance components on all pairs of models.

For leaderboards, Chatbot arena (Chiang et al., 2024) and CRUXEval (Gu et al., 2024) computed confidence intervals using bootstrap with a fixed reference model, which allowed them to obtain per-model confidence intervals but is incorrect (Example 4). Furthermore, Chatbot arena aggregates comparisons with all other models to compute their confidence intervals, which necessarily loses any special properties of a particular pair of models.¹ This work actually compares all pairs of models, and use the summary statistics of all the comparisons to get meaningful noise levels.

Madaan et al. (2024) measured the noise due to random seeds, which is a form of prediction noise in our view. Any direct approach to measure the noise necessarily ignores the data noise (Example 2). For the data noise, they also proposed using unpaired Bernoulli confidence intervals. This method was adopted by Llama 3 (Grattafiori et al., 2024) but is much looser than the paired methods (Example 1) and does not separate the data noise and the prediction noise. Improving model ability is hard and expensive while tightening the analysis is well-established.

6 Discussions

6.1 Remarks on the main findings

Predictable total noise. When the prediction noise is much greater than the data noise, then total noise $\approx$ prediction noise, and prediction noise can be predicted by the Beta theory, so the predictable total noise is a consequence of prediction noise $\gg$ data noise. However, Figure 5 shows non-negligible data noise, and where the total noise still follows the prediction even when the data noise is bigger than the prediction noise at a low temperature². More data is needed to further test if this continue to hold, and if low temperature can reduce prediction noise as the model is training (we guess not, but untested).

prediction noise > data noise. It is not surprising that current LLMs are similar to each other on question level metrics, as they are trained on similar data using similar architecture and learning methods. So it is expected that LLMs perform similarly on each question if they have similar overall abilities. In theory, the prediction noise can still be arbitrarily small, but LLMs seem to be both positively correlated and have sufficient prediction noise for this result to hold.

6.2 Solving hard and special problems

For a problem requiring a long answer where guessing correctly is unlikely, answering even 1 problem might be very significant and interesting. For example, the problem can ask for the proof of an important open problem and the test checks the proof. If a model (or someone) solves such a hard problem then we would not object to the sample size of 1. We probably don’t need to consider this yet. We observe that all pairs of models have enough noisy inconsistencies where model A may beat B on 2 hard questions, but then B beats A on 2 easy questions. If A is so much better than B such that A generated a long and correct solution to an important question, why did it fail on some easy questions where B succeeded? By inspecting the average question level metrics over 1000 samples, we see that worse models also has a smaller chance of guessing the correct answer. For example, if the best model usually answers question x_i correctly with $E_\epsilon[A(x_i)] = 0.9$, then we also see $E_\epsilon[B(x_i)] > 0$ detectable by a mere 1000 samples for a much worse model B .

In our human experience, we intuitively get a lot of signals by evaluating on hard problems with a number of steps (e.g. interviews, math). The key might be to reflect on the details of how the problem was solved. In contrast, LLM evals and RLVR settings only give back 1 bit back after doing a lot of work. If we continue in the direction of smaller and harder evals, we probably need to also generate more information per question, until then, more questions are needed to satisfy regular statistical standard.

6.3 Caveats of the expected metric

With the impressive increase in power, we recommend averaging for most controlled experiments on model architecture, optimization, loss function, etc. that are not expected to have a large effect on the

¹They no longer provide confidence intervals, but the method is described in their paper.

²except for chain-of-thought predictions, where the prediction noise is still higher

eval and do not target the eval specifically. However, there are some caveats if our goal is to improve a particular eval metric using all the best available methods, and when the independent experiment variable is expected to affect the “sharpness” of the distribution. Example 5 shows how improvement in expected accuracy leads to no increase after aggregation, and how sharpening the distribution on good-enough model can lead to much better expected accuracy. Example 6 shows how increased expected accuracy leads to a decrease the final performance when the trade-off is less diversity, which can happen in post-training. Example 7 compares the expected metric vs. alternatives and show that arbitrarily small but consistent changes is significant using the expected metric, clarifying that an independent standard of effect size is also needed.

Example 5 (majority voting). Suppose the expected accuracies on all questions are the same with $E[A] = 0.6$, $E[B] = 0.9$ where the expectation averages out the prediction noise. This big improvement in expected accuracy might be trivially achievable by majority voting if we try to put forward the best prediction instead. Whenever the equivalence of sampled predictions can be checked, majority voting can increase the accuracy to 100% if the equivalence class with the correct answer has the highest probability, which is the case when $E[A] > 0.5$.

Example 6 (reliable verification). Suppose that on half of the questions $E[A] = 0.1$ and $E[B] = 1$, but on the other half of questions $E[A] = 0.1$ and $E[B] = 0$. While $E[B] = 0.5 > E[A] = 0.1$ with B having a much higher average, A is better if the predictions can be verified (search problems or formal proofs) where A is 100% with a verifier but B is only 50%. If a reliable verifier is available, then the *pass-at-large-K* metric is a more meaningful metric than the average.

Example 7 (Expected metric and alternatives). Suppose we have 100 True/False questions, model A has an expected accuracy of 60% and Model B 61% on each question.

Expected metric. We draw enough samples per question to be sure that A is 60% and that B is 61%. **Conclusion:** A and B are significantly different. This metric is used by many papers of the area, and corresponds to the *data noise* after averaging away some prediction noise.

Best prediction metric. Since the probability of correct is greater than half, both models can achieve 100% accuracy if they predict by majority voting or thresholding. **Conclusion:** A is the same as B . This is used by most classification evals and should be used when pushing the eval.

One prediction metric. One sampled prediction per question is evaluated, so A and B do about 60% with a 7% standard error on the difference. **Conclusion:** A is not significantly different from B . This metric is based on a single answer per question, is used in most of the sciences and tests for humans, and corresponds to the *total noise*.

Suppose that on question x_i , $E[A(x_i)] = p_i$, $E[B(x_i)] = p_i + e$, then with enough samples we can conclude that A and B are significantly different no matter how small e is. Still, pure noise without the bias is not expected to produce a significant result.

6.4 Applications and Recommendations

In addition to the recommendation of Miller (2024), we advocate for more “global” noise measurements to decouple noise analysis from most experiments, so practitioners can focus on their experiments instead of the noise analysis. A motivating example for this recommendation is the Llama 3 report, where many tables show the unpaired standard errors, which is too loose and gives no additional information since they are just a simple function of the accuracy. Since some decoupling is possible, eval and leaderboard builders can naturally conduct the noise analysis by running the all-pairs-paired method to measure all the noise types on their eval, thus taking more responsibility for the statistical reliability of their eval. This can even be done by a third-party as in this work, as long as leaderboard maintainers release their question level results like Jimenez et al. (2024); Jain et al. (2024), ideally with multiple predictions per question so the noise components can be better estimated. This might be a unique possibility in the LLM paradigm, where both models and evals have global identities to enable this approach.

For model developers, we offer in increasing accuracy:

- use the heuristics $SE[A - B] \approx SE[A]$ and read these SEs from our tables
- read our paired total noise curves to obtain the ranges of each type of noise. Total noise levels can be used when evaluating using one prediction per question and data noise can be used to determine how effective averaging is.

- use method and estimators to measure the noise components on any data, which enables the detection of much smaller signal in controlled experiments.

If it is verified that prediction noise is dominant, then averaging can be used to greatly increase the power, and the usual method of inferring noise from the training curve can also be justified.

Noise-aware Meta-analysis of multiple evals. We have an increasing number of evals of varying quality and size, so we discuss how to make more out of many evals. There are many huge table of results that are hard to interpret in the literature. Averaging MMLU where $N \sim 10k$ and HumanEval where $N \sim 100$ likely leads to worse results than using MMLU alone. Instead of the absolute improvement, the percent relative improvements, or the confidence intervals, we advocate using the z -score because it is signed and more intuitive, and is well-behaved. Recall the z -score $z = \frac{\bar{A} - \bar{B}}{\text{SE}[\bar{A} - \bar{B}]}$. This work measured the paired and unpaired versions of prediction, data, and total SEs and is suitable here. Inspecting this list of z -scores (one for each eval) and checking their consistency is already informative. To get a meta- z -score given the z -score of eval i , an intuitive method is $\bar{z} = \frac{\sum_i w_i z_i}{(\sum_i w_i^2)^{1/2}}$, where some prior beliefs can be incorporated into w_i . For example, $w_i = 1$ means that we believe each eval has equal weight, whereas $w_i = N_i^{1/2}$ means each question has equal weight. Probably incorporating our prior belief of the importance of each eval into the weight is the best. See Zaykin (2011) for more related work and analysis, on which we have two remarks specific to LLMs: 1) the effect size of doubling the model size (scaling law) can be a reference on the expected effect size on an eval if desired; 2) giving each question from different evals the same weight is less valid for evals since different evals may capture different abilities, even if this has the most statistical power for a pool of iid questions.

6.5 Limitations

While the method is general, all the empirical data are from correctness evals, so whether the empirical findings generalize to other evals remains to be tested. Bowyer et al. (2025) show that CLT does not work well for less than a few hundred questions, though with 100 questions the errors are only significant in the tails. For simplicity, we did not consider the potentially impactful clustered correction of Miller (2024).

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A Analysis details

A.1 Alternative decomposition

Taking expectation over either the noise or the question first has the same total variance and thus the sums are equal

$$\begin{aligned}\text{Var}_{x,\epsilon}[A] &= \text{Var}_x[\mathbb{E}_\epsilon[A]] + \mathbb{E}_x[\text{Var}_\epsilon[A]] \\ &= \text{Var}_\epsilon[\mathbb{E}_x[A]] + \mathbb{E}_\epsilon[\text{Var}_x[A]]\end{aligned}$$

To see that $\mathbb{E}_x[\text{Var}_\epsilon[A(x, \epsilon)]]$ indeed corresponds to the prediction noise that we can measure directly,

$$\text{Var}_\epsilon \left[\frac{1}{N} \sum_{i=1}^N A(x_i, \epsilon_i) \right] = \frac{1}{N^2} \sum_{i=1}^N \text{Var}_\epsilon[A(x_i, \epsilon)] \rightarrow \frac{1}{N} \mathbb{E}_x[\text{Var}_\epsilon[A(x, \epsilon)]]. \quad (3)$$

The left hand side is the prediction noise on N particular data points, which converges to the mean if they are iid. One can directly evaluate the prediction noise on a particular dataset by running the eval K times and measure the variance of the K results. That is also correct in expectation but is more noisy since (3) includes equally valid samples not drawn by any particular eval run.

Paired prediction noise from the same model Using (2) to compute $\text{Var}_{x,\epsilon}[A(x, \epsilon_1) - A(x, \epsilon_2)]$ is too small when using the same set of samples for A . Instead, we can use the following equality and an unbiased estimator for $\mathbb{E}_x[\text{Var}_\epsilon[A]]$ with $A = A(x, \epsilon_1)$, $B = A(x, \epsilon_2)$.

$$\begin{aligned}\text{Var}_{x,\epsilon}[A - B] &= \text{Var}_x[\mathbb{E}_\epsilon[A - B]] + \mathbb{E}_x[\text{Var}_\epsilon[A - B]] \\ &= 0 + \mathbb{E}_x[\text{Var}_\epsilon[A] + \text{Var}_\epsilon[B]] \\ &= 2\mathbb{E}_x[\text{Var}_\epsilon[A]]\end{aligned}$$

Equivalently, we can still use (2) and Table 2 if we sample without replacement when estimating $\mathbb{E}[A(x, \epsilon_1)A(x, \epsilon_2)]$ to avoid the bias over-estimating the covariance using the same set of samples. Suppose we have K samples for a question x with scores $A_1, \dots, A_K$. To estimate $\mathbb{E}[A(x, \epsilon_1)A(x, \epsilon_2)]$ we must use $\text{mean}_{i,j \neq i} A_i A_j = \frac{1}{K^2 - K} \sum_{i,j \neq i} A_i A_j = \frac{1}{K^2 - K} [(\sum_i A_i)^2 - \sum_i A_i^2]$.

B Equivalence to bootstrap and sign test

To setup both methods, let $W_A := \sum_{i=1}^N \mathbb{I}[A(x_i) > B(x_i)]$ be the number of times model A wins against model B and vice versa for W_B , and let's assume $W_A > W_B$ for convenience. The question is how likely we have the same conclusion $W_A > W_B$ under the randomness prescribed by bootstrap or the sign-test.

The bootstrap. We draw another N samples with replacement from the existing samples $x_1, \dots, x_N$ to obtain $X_A, X_B, X_0 \sim \text{multinomial}(N, q_A, q_B, q_0)$ with $X_A + X_B + X_0 = N$, where the outcome probabilities are $q_A = W_A/N$ for A win, $q_B = W_B/N$ for B win, and $q_0 = 1 - q_A - q_B$ for tie. The mean and variance of the resamples are respectively $\mathbb{E}[X_A - X_B] = W_A - W_B > 0$,

$$\begin{aligned}\text{Var}[X_A - X_B] &= \text{Var}[X_A] + \text{Var}[X_B] - 2\text{Cov}[X_A, X_B] \\ &= N(q_A(1 - q_A) + q_B(1 - q_B) + 2q_A q_B) \\ &= N(q_A + q_B - (q_A - q_B)^2) \\ &\approx N(q_A + q_B) = W_A + W_B.\end{aligned} \quad (4)$$

Under bootstrap, a natural question is how likely we would see the opposite result $\Pr[X_A < X_B]$. With an approximation that $(q_A - q_B)^2 \ll q_A + q_B$ (else the result is probably significant already), the bootstrap asks if $\Pr[z \geq \frac{W_A - W_B}{\sqrt{W_A + W_B}}]$ for standard normal z .

Directly estimating the variance from the samples also gives the same answer as (4). While this is a consequence of bootstrap variance equal to the sample variance, a direct calculation is this

$$\begin{aligned}\text{Var}_x[A(x) - B(x)] &= \mathbb{E}[(A - B)^2] - \mathbb{E}[A - B]^2 \\ &= q_A + q_B - (q_A - q_B)^2.\end{aligned}$$

The sign test. When comparing model A vs. model B , the null-hypothesis is that model A and B each has a $1/2$ chance of being better (win) on each question. A wins if A gets a question correct but B does not, tie if A and B are both correct or incorrect. The question is if the observed results are likely to happen under this null-hypothesis.

The p -values is then $\Pr[X \geq W_A]$ for $X \sim \text{binomial}(W_A + W_B, 0.5)$ with $E[X] = \frac{1}{2}(W_A + W_B)$ and $\text{Var}[X] = \frac{1}{4}(W_A + W_B)$. The question is how likely is X to be as extreme as W_A as observed. When $W_A + W_B$ is large, the normal approximation is accurate and reduces to $\Pr[z > \frac{W_A - W_B}{\sqrt{W_A + W_B}}]$ for standard normal z (i.e. one sided). If we also consider how A might be worse, then we should consider the two-sided question. Table 3 shows our variance components in terms of the win rates q_A, q_B used in the sign test.

So the main empirical variance method, the bootstrap, and the sign test give the same answer. A slight approximation was used on the sign test. Without the approximation, $\text{Var}[\text{sign test}] \geq \text{Var}[\text{bootstrap}]$ due to using 0.5 for the null hypothesis instead of the empirical win rate.

Formula	Win rates
$\text{Var}_{x,\epsilon}[A - B]$	$q_A + q_B - (q_A - q_B)^2$
$\text{Var}_x[\text{E}_\epsilon[A - B]]$	$\frac{1}{N} \sum_i q_{A,i} + q_{B,i} - (q_A - q_B)^2$
$\text{E}_x[\text{Var}_\epsilon[A - B]]$	$\frac{1}{N} \sum_i q_{A,i} + q_{B,i} - (q_{A,i} - q_{B,i})^2$

Table 3: For where $A, B \in \{0, 1\}$ with $\Pr[A(x_i) > B(x_i)] = q_{A,i}$, $\Pr[A(x_i) < B(x_i)] = q_{B,i}$.

C Estimator testing

We validate our variance estimators by sampling from generative models where the ground truth is known. Our primary focus is measuring estimation accuracy and how it scales with sample size, though we also check the bias and test some properties as well.

For each test scenario, we generate data $A, B \in \mathbb{R}^{N \times K}$ from models with known variance components, apply our estimators, and compare against ground truth over many independent attempts. We use two primary metrics: (1) root mean square (rms) relative error to measure accuracy, and (2) t -scores to detect systematic bias.

We test under three generative processes. The *Bernoulli model* samples N problems with replacement from a population of size N^* parameterized by $p_A \in \mathbb{R}^{N^*}$, then sample K Bernoulli trials for each of the N problems. The *stratified Bernoulli model* evaluates when every question is sampled exactly once, simulating the case of drawing samples from real eval, breaking independence and introduces tricky effects. The *bootstrap model* resamples from empirical metric matrices, useful for comparisons to bootstrap and testing based on real data.

Accuracy vs. sample size. Our main concern is practical accuracy. Tests establish that estimators achieve rms error below 0.25 for $N = 100$ sample sizes and below 0.13 for larger samples ($N = 400$). We verify this across diverse probability distributions including uniform, Beta, specific distributions and real data. For paired comparisons, tests confirm we get reasonable accuracy (relative rms < 0.1) on the total variance and prediction variance. The data variance is the hardest to estimate since it tend to be the smallest where correcting for the bias in K is critical (Figure 3)..

Bias Testing. Deriving and testing unbiased estimators proved useful for understanding estimator for each type of sampling procedure. The process of eliminating bias forced us to carefully account for all sources of estimation error—finite sample corrections in both N and K , paired sampling effects, and stratified sampling adjustments. Moderately large N is necessary for accurate estimation anyway. We test corrections for N and found they always led to larger errors and thus not recommended. On the other hand, small K is common in practice, so we focus on K , and found it to be more than just a bit of bias correction but critical in the case of large N small K .

This testing builds confidence that our estimators behave as predicted and that we understands how to get unbiased estimators in each situation, even as we only recommend two of the simplest estimators.